Exploring GenAI-Driven Personalized Health Coaching for Chronic Disease Self-Management

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Interim Report

1. Introduction

1.1 Background

Chronic diseases represent a growing burden on healthcare systems worldwide, accounting for nearly three-quarters of deaths globally. In Singapore, approximately one in four adults aged 40 and above lives with at least one chronic condition. Providing continuous, personalised support for these patients outside of clinical settings remains a persistent challenge — one that contributes to poor medication adherence, lifestyle barriers, and reduced patient engagement.

Endometriosis is a chronic inflammatory condition affecting approximately 1 in 10 women globally and around 8% of women in Singapore. It is characterised by heterogeneous, fluctuating symptoms — including chronic pelvic pain, dysmenorrhoea, deep dyspareunia, and cyclic gastrointestinal and urinary symptoms — that fluctuate over weeks and months. Diagnostic delays in the Asia-Pacific region average 6–8 years, driven by cultural normalisation of menstrual pain and inconsistent symptom recognition by both patients and clinicians. Endometriosis is also increasingly recognised as a multi-system condition, associated with elevated risks of cardiovascular disease, autoimmune conditions, and certain cancers.

The current standard of care depends heavily on patient-reported symptoms elicited during brief, infrequent consultations. This model has several well-documented weaknesses. First, patients experience recall bias, struggling to accurately reconstruct weeks of fluctuating symptoms during a 15-minute appointment. Second, patients often find it difficult to articulate subjective experiences such as pain intensity or emotional impact in ways clinicians can act on. Third, fragmented longitudinal data across multiple consultation notes makes it impossible for clinicians to detect trends, gradual escalations, or emerging red flags that would be obvious in continuous data. Fourth, the substantial psychological burden of the disease — including chronic pain, uncertainty, and the experience of being disbelieved — is often unaddressed within clinical workflows.

1.2 Scope and objectives

The project addresses the following research question:

How might we eliminate reliance on slow, error-prone information gathering from patients so that clinicians can make faster and more accurate diagnoses, while simultaneously supporting patients' psychological wellbeing through structured self-reflection?

The purpose of the project is to design and prototype an AI-assisted journalling system that:

1. Enables patients to record symptoms, behaviour, and emotional state in real time, reducing recall bias.
2. Uses Large Language Models (LLMs) to extract structured clinical information from unstructured patient text.
3. Detects clinically meaningful trajectory changes and red-flag symptoms over time.

4. Generates concise, verifiable summary reports for clinicians to review before consultations.
5. Provides therapeutic value to patients through the act of structured self-reflection itself.

The project produces a working end-to-end prototype pipeline that ingests unstructured patient journal entries and outputs one-page clinician summary reports, along with a demonstration interface that allows the pipeline to be shown running on selected patient cases. Evaluation is carried out against a synthetic patient corpus of 30 to 50 simulated patients, using both quantitative metrics (precision, recall, F1 score, sensitivity, specificity) and rubric-based qualitative assessment. Endometriosis serves as the sole target condition for this iteration, chosen so that depth of clinical grounding can take priority over breadth of application within the project timeline.

Several dimensions are explicitly out of scope. No real patient data will be collected, as no ethics approval pathway is available within a JC research timeframe; all data will be either synthetic or paraphrased and anonymised from publicly available sources. The project does not extend to clinical deployment, real-world trials, or partnership with any specific healthcare provider. Wearable sensor integration, while acknowledged as a promising complement to text-based data, is treated as a stretch consideration rather than a built feature — objective signals such as sleep and activity may be simulated within the corpus but will not be captured from actual devices. The system is not intended to provide diagnostic claims or replace clinician judgement; its role is to support consultation efficiency and surface concerning patterns earlier, with all clinical decisions remaining with the treating physician.

2. Methadology

The project follows a mixed methods design combining literature-grounded schema development, synthetic data generation, LLM-based text processing, rule-based clinical analysis, and multi-track evaluation. The team is organised into three parallel work streams — clinical and research foundation, technical pipeline build, and user experience and interface — to maximise parallelisation across the project timeline.

2.1 Data Collection

Due to the constraints of our project, we will use a synthetic patient corpus grounded in three sources:

- Synthetic patient profiles designed by the team and elaborated into daily journal entries with LLM assistance.
- Paraphrased and anonymised excerpts from public patient forums such as Reddit r/endometriosis and Endometriosis UK, never quoted verbatim.
- Case descriptions from peer-reviewed clinical literature reworded into first-person diary form.

The target corpus is 30 to 50 simulated patients, each with 8 to 12 weeks of daily entries. Three trajectory patterns are deliberately seeded into the corpus:

- Stable patients with consistent baseline symptoms.
- Gradually worsening patients with escalating pain and functional impact.

- Patients developing red-flag symptoms during the reporting period.

The deliberately seeded patterns provide ground truth for downstream evaluation of trajectory and red-flag detection.

Over the June Holiday period, we focused on conducting a comprehensive literature review on the different symptoms and disease progression of endometriosis. We narrowed down the symptoms that would affect patients the most, and the most quantifiable ones. We produced 10 entries of over 30 patient profiles, all displaying different symptoms and disease progression. We have also been trying different methods of data analysis and statistical tests to best represent the data, like linear regression or mixed-effects models, non-parametric Friedman test and Pearson correlation, to best represent different data sets and trends.

To incorporate the AI feature, we have been implementing it to comb through the short answer questions and side remarks that the patient makes during weekly entries. It will also be used to look through the data sets and identify anomalies.

2.2 Work done

Two structured interviews were conducted with a patient advocate from EndoSupportSG, providing lived-experience grounding that complements the clinical literature. Several themes emerged that have directly shaped the project design:

- The centrality of medical dismissal and being disbelieved as core patient experiences, reframing the system's purpose as one of helping patients be believed rather than merely producing efficient consultations
- The continuous rather than purely cyclic nature of severe endometriosis symptoms, informing the design of the seeded trajectory patterns in the synthetic corpus
- The importance of accommodating neurodivergent users who may struggle to quantify pain on numerical scales, informing the mixed structured/free-text design of the journalling interface
- The equal weight that psychosocial signals (persistent low mood, hopelessness, expressions of exhaustion) should carry alongside physical red flags in the detection layer

To complement the endometriosis progression monitoring pipeline, we plan to develop an integrated mobile application that serves as both a symptom tracking platform and a comprehensive self-management tool. It will incorporate a menstrual cycle and period tracking feature where users will be able to log menstruation dates, cycle length, bleeding intensity, spotting, and associated symptoms, allowing the system to contextualise symptom progression with hormonal fluctuations throughout the menstrual cycle. This longitudinal data will provide greater insight into symptom patterns and enable more accurate identification of recurring trends, flare-ups, and potential disease progression, and can still be used by users who don't have endometriosis.

The application will also integrate an AI-powered chatbot designed to provide psychological support and improve chronic disease self-management. Living with endometriosis is often associated with increased

levels of stress, anxiety, depression, and emotional fatigue due to persistent pain and uncertainty surrounding symptom progression. The chatbot will provide empathetic conversational support, encourage symptom reflection, answer common questions about endometriosis using evidence-based information, and guide users towards appropriate self-care strategies. It may also recommend relaxation techniques, coping strategies, or prompt users to seek professional medical or psychological assistance when emotional or physical symptoms are identified. Where appropriate, the chatbot can encourage continued symptom logging and medication adherence, promoting greater engagement with the monitoring platform.

Substantial progress has been made on the user experience and interface stream. We produced a full set of high-fidelity screen designs for the intended system. From these designs, we then generated a functional front-end prototype in code, giving us a working interface that demonstrates the intended user flow even while the underlying analytical pipeline is still under development.

Five core screens have been designed and prototyped:

1. Patient Dashboard
    a. A home view showing today's pain level and mood at a glance, cycle overview with current phase and days to ovulation, recent symptoms, a weekly pain trend chart, and a wellness reminder for medication.
2. Daily Symptom Log
    a. the primary journalling interface, combining a pain intensity slider (0–10), pain location chips (pelvic, abdominal, back, legs), mood selection chips, a free-text journal entry field, and medication logging.
3. AI Health Insights
    a. a patient-facing view of the AI-generated insights, including a weekly summary, pattern recognition showing symptom-cycle correlations, personalised recommendations, and a conversational emotional support component ("Lumina").
4. Clinician Clinical Summary
    a. the doctor-facing pre-consultation summary, showing patient identifiers, a symptom trend chart, key symptoms, current medications, notes, and flagged concerns requiring attention.
5. Patient Profile
    a. standard settings and personal information, including medical history, notification preferences, and language toggle.

The prototype at this stage is a working front-end demonstrating the intended user experience, it does not yet integrate with the pipeline. Its function at this stage is to demonstrate the intended user experience

3. Future work

The next phase of this project will focus on expanding, integrating, and evaluating the proposed endometriosis monitoring system. Building on the literature review and the development of a synthetic dataset, we will refine our statistical analyses to identify the most clinically meaningful methods for modelling disease progression. We will also continue improving the AI pipeline by enhancing symptom

extraction from free-text journal entries, strengthening anomaly detection, and testing the system against challenging scenarios such as ambiguous responses, false positives, and missed medication records to ensure reliability and minimize hallucinations.

In addition, we plan to develop an integrated mobile application that combines symptom tracking with a menstrual cycle and period tracker, allowing users to record cycle characteristics alongside endometriosis symptoms for more comprehensive longitudinal monitoring. Finally, the complete four-stage pipeline will be evaluated as an end-to-end system. We will assess the accuracy of symptom extraction, trajectory analysis, red-flag detection, and clinician summary generation using quantitative performance metrics and manual validation. The usability and clinical relevance of the generated summaries will also be evaluated to ensure the platform is transparent, reliable, and capable of supporting both patients and healthcare professionals in the long-term management of endometriosis.

5 Conclusion

At this stage, the project is still in its early stages of development, with our work primarily focused on conducting an extensive literature review, brainstorming potential solutions, designing the system architecture, and planning the methodology. These preliminary efforts have established a strong foundation by identifying clinically relevant symptoms, designing the AI-assisted monitoring pipeline, and outlining the key features of the proposed application.
Following this, we will dedicate our efforts to implementing and rigorously evaluating each component of the system. We also intend to seek feedback from clinicians and other healthcare professionals to assess the clinical relevance, usability, and practicality of the proposed dashboard and AI-generated summaries. Their input will be used to refine the system to better support real-world clinical workflows and patient care.
We aim to produce a robust, transparent, and clinically meaningful platform that not only demonstrates technical innovation but also addresses the practical needs of both patients living with endometriosis and the healthcare professionals who care for them.

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